



**UTM**  
UNIVERSITI TEKNOLOGI MALAYSIA

DEPARTMENT OF COMPUTER SCIENCE

FACULTY OF COMPUTING

UNIVERSITI TEKNOLOGI MALAYSIA

**SECP3106-02 - APPLICATION DEVELOPMENT**

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**PROPOSAL**

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**GENIUSAQILOS: An Integrated Teacher Management System for Tadika Aqil Miqail  
Using Dart-Based Application Framework**

**Group: Hackatoons**

| NAME                              | MATRIC NO |
|-----------------------------------|-----------|
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## Problem Background

The rapid advancement of digital technology necessitates that educational institutions, including early childhood centers like Tadika Aqil Miqail, optimize their human resource management to enhance operational efficiency. Currently, managing teacher data, monitoring daily assignments, and tracking performance are often handled through conventional methods, which pose risks of data inconsistency and hinder real-time decision-making. Consequently, **GENIUSAQILOS** is developed as a Dart-based integrated solution designed to digitalize all aspects of faculty management, providing a structured framework ranging from secure user authentication to comprehensive teacher record-keeping.

The application integrates critical modules such as leave management, task and duty tracking, and teaching trackers to ensure that every teacher's responsibilities and attendance are accurately monitored. Furthermore, the performance tracker and reporting system within GENIUSAQILOS enable the management of Tadika Aqil Miqail to conduct objective evaluations of teaching effectiveness on a regular basis. By implementing this system, the administrative burden is expected to be significantly reduced, allowing the institution to shift its primary focus toward improving the quality of education and learning outcomes for the students at Tadika Aqil Miqail.

## Objective

## Financial Budgeting

## Target Users

The application that will be developed are intended to be used by:

1. **Principal:** The principal who will monitor the overall performance of each teacher, approve leave applications, assign duties and tasks, track teachers' attendance and activities, and also generate reports.
2. **Teachers:** Teachers can update their profiles, apply for leave, view and get notified for assigned tasks and duties, and submit training documents.
3. **Administrators:** Admins are involved in facility maintenance, which includes obtaining reports and complaints from teachers using the system. Admins are also responsible for teachers' records and disciplinary actions.

## Mandatory Modules

The mandatory module comprises Login, Register, and Manage User modules. Each components of the modules are explained in detail below :

- **Login**
  - a) **Multi-Role Entry** - Ensure secure login and accessibility for different roles automatically, which are the teachers, administrators, and principal.
  - b) **Security** - Password encryption and password recovery via registered email.
  - c) **Session Management** - "Remember Me" button to keep users logged in on their personal mobile devices for convenience.
- **Register**
  - a) **Application Submission** - New teachers fill in their basic details such as ----- and upload mandatory digital documents -----.
  - b) **Notify Account Status** - Notify account status update from “Pending” to “Verified” after account verification from administrators.
  - c) **Verification** - Input verification to ensure no errors in formatting and no duplicate IC numbers or emails are used, and using eKYC to ensure the entered IC numbers match the document uploaded.
- **Manage User**
  - a) **View User Information** - The admin will be able to view the list of all the teachers working in the preschool along with their information.
  - b) **User Application Authorisation** - This feature enables the admin to approve or reject the user application after validating the documents required.
  - c) **Edit User Information** - This feature enables the admin to edit any information about the teachers if necessary.
  - d) **Deactivate/Delete User** - This feature allows the admin to deactivate the user if they are on a break or delete the user information if they have stopped working in the particular preschool.
  - e) **Role Management** - This feature allows the admin to add a new admin role if needed.

## Industry Modules

The industry module comprises Teacher Records, Leave Management, Task & Duty Manager, Teacher Tracking Tracker, Teacher Performance Tracker and Teacher Reporting System modules. Each components of the modules are explained in detail below :

- **Teacher Records**

- a) Display essential teacher information such as full names, IC numbers, and contact details.
- b) Provides a secure place to upload digital copies of MyKads, resumes, and academic certificates.
- c) Keeps a record of emergency contact persons and medical check-up reports for every staff member.

- **Leave Management**

- a) Leave Application for various leave types, including Birthday Leave redemption. (tell 10 type of leave and
- b) Real-Time Leave Tracker to track leave balance and record leave data.
- c) Leave approval or rejection requests by admins while monitoring minimum staffing limits to prevent shortages.
- d) Notify teachers regarding approval status, rejections, appeals, and also remarks on rejected leave, such as missing MC documents or .

- **Task & Duty Manager**

- a) Assign tasks fairly to teachers which are cleaning duty, arrival duty, dismissal duty, half-full day transition duty and assembly duty.
- b) Auto-generate duty schedules
- c) Provide clear checklist for teacher's task
- d) Track task completion
- e) Send reminder/notification for each task

- **Teacher Training Tracker**

- a) Allows teachers to record the title, date, and location of any professional courses they attended.
- b) Enables teachers to upload certificates and photos as evidence of their participation.
- c) Helps track if teachers are meeting the requirement of attending at least three training sessions per year.
- d) Includes a section for teachers to write about what they learned and how it helps their teaching.

- e) Summarizes all training data and participation history into a simple report for yearly staff reviews.
- **Teacher Performance Tracker**
  - a) Allow administrators to evaluate teacher KPIs using ten specific categories, such as attendance, classroom management, and teaching performance, to track how well teachers are doing.
  - b) Allow administrators to monitor classroom standards by checking if classrooms are clean, organized, and if safety rules are being followed correctly.
  - c) Allow administrators to track student development by recording how well teachers are helping students progress, keeping them disciplined, and encouraging them in class.
  - d) Allow administrators to assess professionalism by evaluating how teachers communicate with parents and colleagues, and if they follow the school's dress code and SOPs.
  - e) Allow administrators to measure initiative and growth by tracking if teachers are attending required trainings and taking the initiative to create new activities or decorate classrooms.
- **Teacher Reporting System**
  - a) Allow teachers to submit reports regarding workplace conflicts such as sexual harassment, bullying and conflict between the staff.
  - b) Enables teachers to report work-related issues such as SOP violations, workload stress and teacher misconduct.
  - c) Allow teachers to report on the facility and classroom-related problems such as facility maintenance, teaching material shortages and safety hazards.
  - d) Enables teachers to do a report regarding the IT or system problem.
  - e) Make the reports visible to the principal for monitoring and further action.

### Module Distribution Table

| Module    | Person In Charge (PIC) |
|-----------|------------------------|
| Main Page | Nurul Asyikin          |
| Login     | Anis Safiyya           |
| Register  | Anis Safiyya           |

|                             |               |
|-----------------------------|---------------|
| Manage User                 | Harini        |
| Teacher Records             | Haani         |
| Leave Management            | Anis Safiyya  |
| Task & Duty Manager         | Harini        |
| Teacher Training Tracker    | Haani         |
| Teacher Performance Tracker | Jeanette      |
| Teacher Reporting System    | Nurul Asyikin |

### **Justification**

The modules were divided among the group members according to each module's complexity, personal preferences and each member's ability to handle the assigned task. Anis Safiyya was assigned to handle the Login, Register and Leave Management modules which are all linked to user management and access. Harini was in charge of the Manage User, as well as the Task and Duty Manager modules as both modules involve administrative management features.

Next, Haani was assigned to deal with the Teacher Records and Teacher Training Tracker modules as both modules are related to teachers' data and progress tracking. Jeanette was assigned the Teacher Performance Tracker module as a solo task due to the higher complexity and detailed functionality required for the performance evaluation. Lastly, Nurul Asyikin was assigned the Main Page and Teacher Reporting System modules to handle general user interface and report management functionality. This module distribution helps ensure the workload is balanced among all members.

### **Conclusion**

By the end of this semester, our team aims to deliver a fully integrated Teacher Management System (GeniusAqilOS) that streamlines administrative tasks from leave management to performance tracking. Our goal is to reduce the manual workload of the administrative and educators at Tadika Aqil Miqail with a centralized platform to ensure strict SOP compliance, and ultimately enhance the quality of early childhood education through better organized teaching and training resources.